

The research of brain-computer interface based on AAR parameters and neural networks classifier

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Abstract—The brain-computer interface(BCI) based on motor imagery was investigated in this paper. A neural networks classifier was adopted to solve the problem of lower classification accuracy in BCI. Firstly, mu rhythm EEG was obtained with a bandpass filter from the subject's scalp electroencephalography (EEG). Secondly, the Kalman Filter algorithm was used to build the adaptive autoregressive model from EEG. The model parameters were used as features of EEG. Lastly, the AAR feature parameters were classified by the neural networks classifier. A compare on the performance between the neural networks and linear discriminant analysis(LDA) was conduct in the simulation. The results show the performance of neural networks is higher than linear discriminant analysis.

Keywords-brain-computer interface; motor imagery; adaptive autoregressive model; neural networks

I. INTRODUCTION

A brain-computer interface is a system in which an individual sends commands to the external device by generating specific patterns in brain signals and a computer detects the brain signal patterns and translates them into commands that identify the individual's intention [1, 2]. The major goal of BCI research is to develop systems that make it possible for disabled users to communicate with other persons, to control artificial limbs, or to control their environment. It has high value both in theory and application area of rehabilitation [3].

Based on the recording methods of brain signals, BCI can be categorized into of invasive and non-invasive BCI. The non-invasive BCI is based on a noninvasive method to measure brain-activity, namely the electroencephalogram (EEG). This BCI is safe and convenient. It can be accepted by most patients. According to the types of input signals, BCI can be divided into BCI based on P300, BCI based on visual evoked potential, BCI based on motor imagery and BCI based on slow cortical potentials. Among them, EEG signals of BCI based on motor imagery is produced only through sensorimotor activity. It do not need any sensation stimulus. So it is BCI in the real sense [4]. Feature extraction and pattern discrimination are essential steps in the BCI. Features extraction methods in BCI base on motor imagery have power spectrum [5-7], AR model parameters [8,9], band power [10] and combination of different features extraction methods etc. Feature classification methods

in BCI base on motor imagery include LDA, quadratic classifier, neural networks and support vector machine etc [11-13].

It is well-known that EEG signals, particularly in EEG based BCI systems, are weak signals of microvolt level. They are also non-stationary. The non-stationarities may be caused by the capricious feedback, changes in electrode impedance or dynamically changing environments. The AAR model can describe the non-stationarities in the EEG. So AAR model parameters is used as feature of left and right hand motor imagery in this paper. The Kalman filter algorithm was used to evaluate the parameters of AAR model. Neural networks classifier is used to discriminate different imagery tasks. In simulation experiments, the performance of neural networks classifier and LDA is compared.

II. EXPERIMENT DATA

This dataset was recorded from a healthy subject (25 years old, female) during a feedback session [14]. The subject sat in a relaxing chair with armrests. The task was to control a feedback bar by means of imagery left or right hand movements. The order of left and right cues was random. The experiment consists of 7 runs with 40 trials each. All runs were conducted on the same day with several minutes break in between. Given are 280 trials of 9s length. The first 2s was quite, at $t=2s$ an acoustic stimulus indicates the beginning of the trial, and a cross was displayed for 1s; then at $t=3s$, an arrow (left or right) was displayed as cue. At the same time the subject was asked to move a bar into the direction of according to the cue. The feedback was based on AAR parameters of channel C3 and C4, the AAR parameters were combined with a discriminant analysis into one output parameter. The recording was made using a G.tec amplifier and a Ag/AgCl electrodes. Three bipolar EEG channels (anterior '+', posterior '-') were measured over C3, Cz and C4. The EEG was sampled with 128Hz, it was filtered between 0.5 and 30Hz. The trials for training and testing were randomly selected. This should prevent any systematic effect due to the feedback.

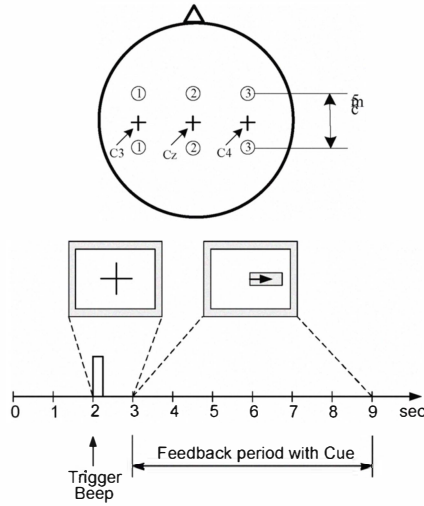


Figure 1. Electrode positions and timing scheme

III. FEATURE EXTRACTION

A. AAR Model

An autoregressive model can be used to describe the EEG in the stationary case. In order to consider the non-stationarity, the model parameters of AR are allowed to vary in time. The AAR model is described by the following equation (1).

$$y_k = a_{1,k}y_{k-1} + a_{2,k}y_{k-2} + \dots + a_{p,k}y_{k-p} + v_k \quad (1)$$

v_k is a zero-mean-Gaussian-noise process with variance $\sigma_v^2(k)$. The index k is usually an integer and describes discrete, equidistant time points. y_{k-i} with $i=1 \dots p$ are the p previous sample values. p is the order of the AAR model. a_{ik} are the time-varying AAR model parameters. The difference to the conventional AR model is that the AAR parameters is allowed to vary in time. This makes AAR model can describe non-stationarity of EEG data.

B. AAR Estimation Algorithm Base on Klman Filter

There are several parameter estimation methods for AAR model such as Least Mean Squares algorithm, the Recursive Least Squares algorithm, Recursive AR algorithm and Kalman Filter. In this paper AAR parameters are estimated with Kalman Filter [15]. The method has higher time resolution and need little calculation. It is appropriate for EEG on-line analysis. Kalman Filter is an algorithm for estimating the state vector of a state space model through an iteration process. AAR model (1) can be rewritten as state equation (2) and observation equation (3).

$$\mathbf{a}_k = \mathbf{a}_{k-1} + \mathbf{w}_k \quad (2)$$

$$y_k = \mathbf{Y}_{k-1}^T \mathbf{a}_k + v_k \quad (3)$$

$$\mathbf{a}_k = [a_{1,k}, \dots, a_{p,k}]^T, \quad \mathbf{Y}_{k-1} = [y_{k-1}, \dots, y_{k-p}]^T. \quad \mathbf{w}_k \text{ is a}$$

multivariate zero mean random noise with covariance $\mathbf{W}_k$. v_k is the measurement noise with zero mean and variance V_k . Kalman filtering is a method that estimates the state $\mathbf{a}_k$ of the system from measuring the output signal y_k .

$$e_k = y_k - \mathbf{Y}_{k-1}^T \mathbf{a}_{k-1} \quad (4)$$

$$(\mathbf{A}\mathbf{Y})_{k-1} = \mathbf{A}_{k-1} \mathbf{Y}_{k-1} \quad (5)$$

$$\mathbf{Q}_k = \mathbf{Y}_{k-1}^T \mathbf{A}_{k-1} \mathbf{Y}_{k-1} + V_k \quad (6)$$

$$\mathbf{K}_k = \mathbf{A}_{k-1} \mathbf{Y}_{k-1} / \mathbf{Q}_k \quad (7)$$

$$\hat{\mathbf{a}}_k = \mathbf{a}_{k-1} + \mathbf{K}_k^T e_k \quad (8)$$

$$\mathbf{Z}_k = \mathbf{A}_{k-1} - \mathbf{K}_k \mathbf{Y}_{k-1}^T \mathbf{A}_{k-1} \quad (9)$$

$$\mathbf{A}_k = \mathbf{Z}_k + \mathbf{W}_k \quad (10)$$

e_k is the one-step prediction error. $\mathbf{A}_k$ is the priori state error correlation matrix. $\mathbf{Q}_k$ is the prediction variance. $\mathbf{K}_k$ is the Kalman gain. $\mathbf{Z}_k$ is the posteriori state error correlation matrix.

In equation (6), equation (7) and equation (9), $(\mathbf{A}\mathbf{Y})_{k-1}$ is used repeatedly. So it is calculated separately and stored to improve the speed of the calculation. The initial value $\mathbf{a}_0$ of $\mathbf{a}_k$ is zero vector. The initial value $\mathbf{A}_0$ of $\mathbf{A}_k$ is the identity matrix of p orders.

In this paper, AAR is used to model the EEG data of channel C3 and channel C4. Though the simulations of different model orders, 6 is chosen as AAR model order. The 6 order model can gain lower computational effort and higher resolution. The values of measurement variance V_k and the observation covariance $\mathbf{W}_k$ can refer to tables 1 and 2 of bibliography [15]. In this paper V_k is calculated with the equation (12). In equation (12), UC is the update coefficient and is set to be 0.001. $\mathbf{W}_k$ is calculated with equation (12) and (13).

$$V_k = (1 - UC)V_{k-1} + UCe_k^2 \quad (11)$$

$$\mathbf{Z}_k = (\mathbf{I} - \mathbf{K}_k \mathbf{Y}_{k-1}^T) \mathbf{A}_{k-1} \quad (12)$$

$$\mathbf{W}_k = UC^2 \mathbf{I} \quad (13)$$

IV. NEURAL NETWORK CLASSIFIER

NN is a nonlinear system composed of many neurons. The neural networks can simulate human brain neural system, such as information processing, storage and searching. Thus NN has some intelligent functions. It has been widely used in nonlinear mapping, time-series estimation and classification. In this paper NN is used as a classifier to discriminate the motor imagery

task of left hand and right hand. Higher classification performance is obtained.

Fig. 2 is the diagram of a forward networks architecture of three layers. It is composed of many neurons, which are connected through the weights. All weights (including bias of all neurons) must be modified before NN is able to be used. The forward network carries out a static mapping from input signals to outputs. There are vast majority of NN weights training methods among which the BP is the most famous. In the standard BP algorithm, weights are modified by utilizing first-order derivative information of errors. Weights are updated along the negative gradient of the cost function. However, convergence to poor local optima and lower learning velocity confine the application of these algorithms. Some enhanced training methods have been introduced by many researchers such as BP algorithm with additional momentum and variable metric method etc.

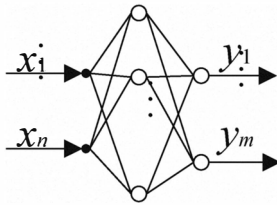


Figure 2. The diagram of a forward network architecture

In this paper a three layers NN was used to solve a two classification problem in BCI based left hand and right hand motor imagery. As a non-linear classifier, NN has a higher classification resolution. The input number is twelve, the output neuron number are two and the number of the hidden neurons are thirty. Sigmoidal function is used as the hidden layer neurons and the output layer neurons activation function. The initial values of the weights are the random numbers between -1 and 1. The input values of NN is the AAR model parameters of C3 and C4 channels. There are 12 parameters of the two channels. The target outputs of the two classes are [1, 0] and [0, 1], respectively indicating the left hand and right hand motor imagery. The training method of NN is the improved BP algorithm with additional momentum. The method can increase the study velocity of NN.

V. SIMULATION RESULTS

LDA classifier was firstly used to discriminate the AAR features of EEG data. The classification result is showed in Fig. 3. The thick line shows the time course of the cross validated error rate, calculated 8 times per second; the thin line displays the error rate for each sample time (128 per second) without cross-validation. Fig. 4 is the average output of LDA of 8 times classification per second in Fig. 3. The thick line is the average values of LDA output of left hand motor imagery. Whereas, the dotted line is the average values of LDA output of right hand motor imagery. From Fig. 3 and Fig. 4, we can conclude that LDA can correctly discriminate the features of the left hand and right hand motor imagery, but the classification accuracy rate is not high.

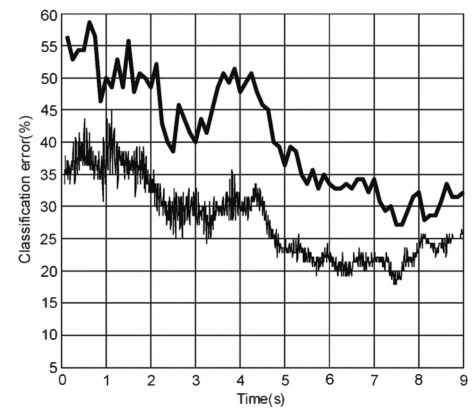


Figure 3. The classification results of LDA

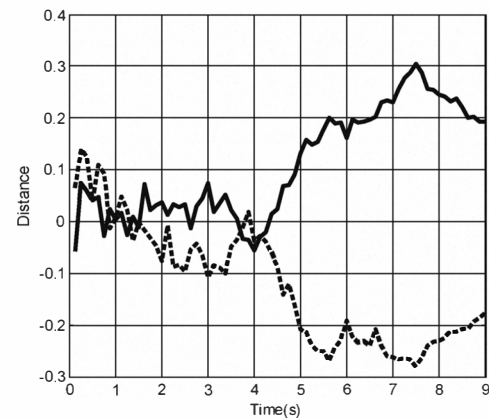


Figure 4. The averaged distance of the output of LDA

Fig. 5 is the comparison diagram of the classification results of LDA and NN. The thick line is the classification error of NN. The dotted line is the classification error of LDA. Both are the classification results of the fourth cross validate, 8 times classification per second. The classification of Fig. 5 is from the fourth second. This is because the subject begin the motor imagery from the third second. The feature of motor imagery is obvious from the fourth second. We can conclude from Fig. 5 that the classification error of NN is lower than that of LDA. The classification accuracy of NN is higher than LDA. Fig. 6 is the training error of NN.

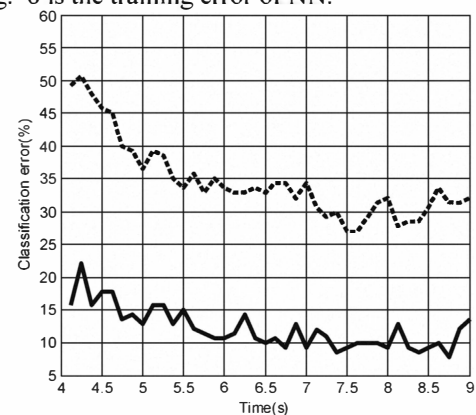


Figure 5. The classification results of NN

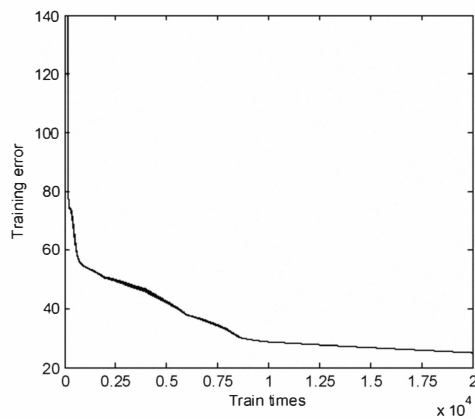


Figure 6. The curve of error of network training

VI. CONCLUSION

The AAR model parameters was used as the feature of different motor imagery tasks. The BP NN with additional momentum was used to discriminate the feature parameters of C3 and C4 channel. An comparison is conducted about the classification performance of NN and LDA in simulation experiment. The simulation results show that the classification accuracy of NN is higher than the LDA.

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